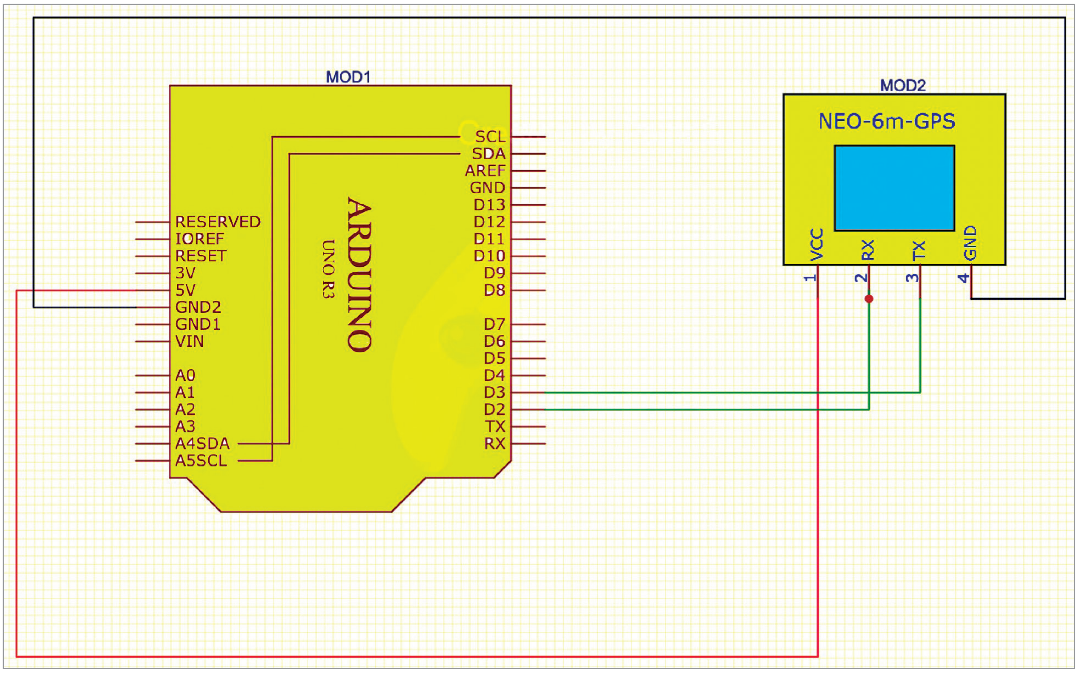

Fig. 4: GPS tracker circuit

```
#include <SoftwareSerial.h>

// Define the RX and TX pins for the SoftwareSerial interface
#define GPS_TX_PIN 2
#define GPS_RX_PIN 3

// Create a SoftwareSerial object for communication with the NEO-6M GPS module
SoftwareSerial gpsSerial(GPS_RX_PIN, GPS_TX_PIN);

void setup() {
  // Initialize serial communication with the computer
  Serial.begin(9600);

  // Initialize serial communication with the GPS module
  gpsSerial.begin(9600);
}

void loop() {
  // Read data from the GPS module and send it to the computer
  if (gpsSerial.available()) {
    char c = gpsSerial.read();
    Serial.write(c);
  }

  // Read data from the computer and send it to the GPS module
  if (Serial.available()) {
    char c = Serial.read();
    gpsSerial.write(c);
  }
}
```

Fig. 5: Raw data code snippet

ate interactive projects. The heart of the Arduino Uno is the ATmega328P microcontroller, which is responsible for executing the program and controlling the project. It has 14 digital input/output pins. It runs at 16MHz clock speed and has a USB connection, allowing the code to be uploaded from the computer to the board and communicated with.

The NEO-6M GPS module. It is a compact, low-cost GPS receiver module widely used in various electronic applications that require accurate positioning and timing information. An introduction to the NEO-6M GPS module along with its specifications can be seen in Fig. 3.

Circuit diagram

Fig. 4 shows the circuit diagram of the GPS tracker. It is built around the Arduino Uno (MOD1) and NEO-6M GPS module (MOD2). Pins D2 and D3 of the Arduino Uno board are connected to pins 2 (RX) and 3 (TX) of the NEO-6M GPS module, respectively.

As the GPS module works on the serial peripherals, it can be attached to any software or hardware pin on Arduino. Here, in this design, the software serial pin is used.

The GPS module is powered with the Arduino 5V pin and GND pin, and Arduino is powered from USB on Arduino.

Software

The code is created in Arduino IDE. The GPS gives serial data so, here the SoftwareSerial library has been used. Set the RX and TX pins for the SoftwareSerial interface. The baud rate is set as 9600, and next, the